

# NIMBUS-9<sup>3</sup>/<sub>4</sub>

## HARDWARE FULL-STACK — DRAWING PACKAGE

Real Quidditch · broom + three balls · BOM · OpenSCAD CAD · drawings

SHEET 1    **BROOM**            manned eVTOL, 8 internal ducted fans

SHEET 2    GOLDEN SNITCH evasion drone, real flapping wings

SHEET 3    BLUDGER            pursuit + soft-tag, zero contact

SHEET 4    QUAFFLE            catch / carry / throw / score

SHEET 5    MATCH FLEET    rollup cost for one 7v7 match

*"Magic? No. We just wrote the collision-avoidance really well."*

NIMBUS-9¾ BROOM

Manned eVTOL — 1 rider, 8 ducted fans, fly-by-intent

Project NIMBUS-9¾ · Real Quidditch · hardware full-stack

ISOMETRIC

SIDE

TOP

CUTAWAY — internal drive revealed

KEY DIMENSIONS & SPEC

(every number ties back to the flight software)

Overall length2400 mm

core/params.py LEN = 2.4 m

Lift fans8 × EDF (internal)

\_default\_fan\_layout (exact x/y)

Thrust-to-weight2.45 (hover @ 41%)

analysis/flight\_thermal.py

Battery sprint~61 s / charge

2600 Wh alone — a sprinter

Match endurance~26 min (hybrid)

series-hybrid: analysis/endurance\_match.py

Cooling margin8.1× waste heat

lift air = coolant + heat pipes

League ceiling150 m (svc ~km)

altitude\_ceiling.py + fence\_ceiling

Control inputwave-to-fly (gesture)

firmware/rider\_input.py -> intent mapper

NOTE

Disc area is capped by what fits inside the bristle flare (the fan never protrudes), so battery-only it's a ~60 s sprinter. A slim in-shaft SERIES-HYBRID genset burning SAF lifts that to a ~26 min match (the battery becomes the peak buffer, full ~250 kW bursts kept). Fans

internal, lift air = coolant. And you fly it by WAVING its handle, pose -> RiderIntent -> the existing intent mapper. FLIGHT + THERMAL + MATCH all PASS.

BILL OF MATERIALS — BRM (41 line items, prototype qty 1)

| REF    | PART                             | SPEC                                                             | QTY | \$ EA | \$ EXT |
|--------|----------------------------------|------------------------------------------------------------------|-----|-------|--------|
| BRM-01 | Carbon-fibre monocoque body p... | 2.4 m, hides the fan bay; SI->mm x1000 in cad/broom.scad         | 1   | 2,200 | 2,200  |
| BRM-02 | Handle shaft + brass pommel      | tapered Ø34->Ø22 mm, leather-wrapped grip                        | 1   | 180   | 180    |
| BRM-03 | Bristle fairing leaf             | ~780 carbon-straw strands, aerodynamic shroud                    | 1   | 640   | 640    |
| BRM-04 | Electric ducted fan (EDF) unit   | 360 N max each; main-lift in bristle shroud + in-shaft trim fans | 8   | 520   | 4,160  |
| BRM-05 | Brushless motor for EDF          | ~12 kW peak, 6S-14S, duct-matched                                | 8   | 340   | 2,720  |
| BRM-06 | ESC (electronic speed control... | 200 A, telemetry, 1st-order ~40 ms response                      | 8   | 150   | 1,200  |
| BRM-07 | Li-ion buffer battery pack       | 2600 Wh, serves the ~250 kW peaks in the hybrid                  | 1   | 3,100 | 3,100  |
| BRM-08 | Battery management system (BM... | 14S smart BMS, cell balancing + SoC telemetry                    | 1   | 260   | 260    |
| BRM-09 | Power distribution board         | 800 A bus, redundant rails                                       | 1   | 180   | 180    |
| BRM-10 | Flight controller MCU            | Cortex-M4F (thumbv7em-none-eabihf), 400 Hz loop                  | 1   | 95    | 95     |
| BRM-11 | IMU (redundant)                  | triple-redundant 6-DOF, 1 kHz                                    | 3   | 55    | 165    |
| BRM-12 | Magnetometer                     | 3-axis, maneuver-compensated                                     | 2   | 18    | 36     |
| BRM-13 | Barometer                        | high-res, for altitude hold                                      | 2   | 12    | 24     |
| BRM-14 | RTK-GPS receiver + antenna       | cm-grade, dual-band                                              | 1   | 420   | 420    |
| BRM-15 | UWB ranging tag                  | indoor cm positioning to pitch anchors                           | 2   | 40    | 80     |
| BRM-16 | Solid-state LiDAR                | 360deg, edge collision avoidance                                 | 2   | 600   | 1,200  |
| BRM-17 | Low-latency datalink radio       | redundant 5G + dedicated mesh                                    | 2   | 140   | 280    |
| BRM-18 | Ballistic recovery parachute ... | Ø7.6 m canopy, rocket-deployed, full-altitude envelope           | 1   | 2,200 | 2,200  |
| BRM-19 | Airbag skirt                     | underbody impact airbag, impact-triggered                        | 1   | 450   | 450    |
| BRM-20 | Master kill-switch receiver      | referee e-stop -> gentle descent                                 | 1   | 60    | 60     |
| BRM-21 | 5-point harness + saddle         | straddle seat, energy-absorbing                                  | 1   | 320   | 320    |
| BRM-22 | Rider intent controls            | handlebar grips, throttle, lean sensors                          | 1   | 210   | 210    |
| BRM-23 | Brass compass/attitude gauge     | cosmetic + real attitude readout                                 | 1   | 90    | 90     |
| BRM-24 | LED trajectory strip             | addressable, night-match light show                              | 1   | 70    | 70     |
| BRM-25 | Heat pipe (sintered copper)      | Ø6 mm, full-shaft length, motor->shaft spreader                  | 4   | 22    | 88     |
| BRM-26 | Air-over-motor duct shroud       | routes lift airflow across windings (air = coolant)              | 8   | 16    | 128    |
| BRM-27 | In-duct winding temp sensor      | NTC, one per fan, feeds the governor                             | 8   | 3     | 24     |
| BRM-28 | Thermal interface material       | phase-change pad, motor + ESC to spreader                        | 1   | 18    | 18     |
| BRM-29 | DShot ESC wiring harness         | bidirectional DShot600, 8 channels + telem                       | 1   | 45    | 45     |
| BRM-30 | CAN-FD vehicle bus               | 5 Mbit, BMS + ESC telem + pose to Dementor                       | 1   | 30    | 30     |
| BRM-31 | Hardware abstraction layer       | firmware skin around the Rust core                               | 1   | 0     | 0      |
| BRM-32 | Micro-turbine genset             | ~125 kW continuous, slim in-shaft series-hybrid range extender   | 1   | 9,800 | 9,800  |
| BRM-33 | FADEC + start system             | full-authority turbine control + spin-up                         | 1   | 1,200 | 1,200  |
| BRM-34 | SAF fuel bladder + lines         | ~18 L crash-safe bladder inside the shaft                        | 1   | 340   | 340    |
| BRM-35 | Sustainable aviation fuel        | 15 kg SAF per fill (~48x li-ion energy density)                  | 1   | 30    | 30     |
| BRM-36 | Exhaust ejector + heat exchan... | entrains bypass air, drops exhaust <150 C                        | 1   | 420   | 420    |
| BRM-37 | Ceramic-matrix bristle tips      | high-temp bristles at the exhaust path                           | 1   | 260   | 260    |
| BRM-38 | Gesture input transducer         | handle IMU + twist grip -> RiderIntent                           | 1   | 0     | 0      |
| BRM-39 | Air-data system                  | pitot + density/OAT probe for altitude compensation              | 1   | 180   | 180    |
| BRM-40 | Reserve parachute                | second canopy, independent deploy                                | 1   | 1,400 | 1,400  |
| BRM-41 | High-ceiling league profile      | fence_ceiling = 150 m (vs 18 m demo default)                     | 1   | 0     | 0      |

UNIT TOTAL (prototype):

\$34,303

DRAWING

NIMBUS-9¾

2026-06-26

UNITS

mm / SI

SCALE

NTS

PART

BRM

REV

v0.1

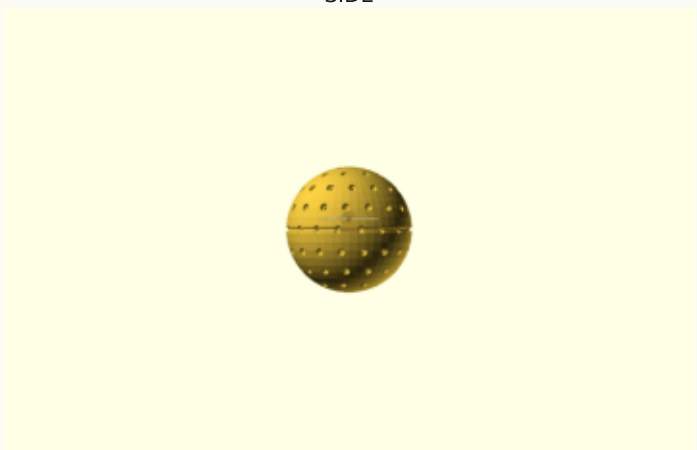
CAD

cad/broom.scad

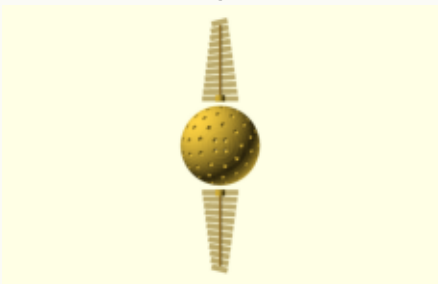
ISOMETRIC



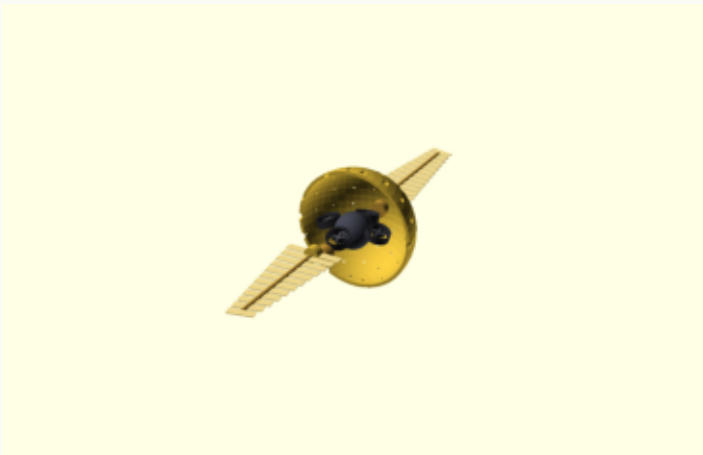
SIDE



TOP



CUTAWAY — internal drive revealed



KEY DIMENSIONS & SPEC

(every number ties back to the flight software)

Shell radius  
ball/params.py snitch.radius = 0.04 m  
40 mm (Ø80)

Mass target  
snitch.mass = 0.05 kg  
50 g

Top speed  
snitch.max\_speed  
19 m/s

Max accel  
snitch.max\_accel (absurdly agile)  
32 m/s²

Lift  
swarm core – the real flight  
4 × micro-EDF (internal)

Wings  
coreless gearmotor driven  
2 × flapping, 8-14 Hz

Capture  
capture\_radius 0.22 / capture\_dwell  
hand held 0.25 s

Danger sense  
snitch.danger\_radius  
4.0 m

NOTE

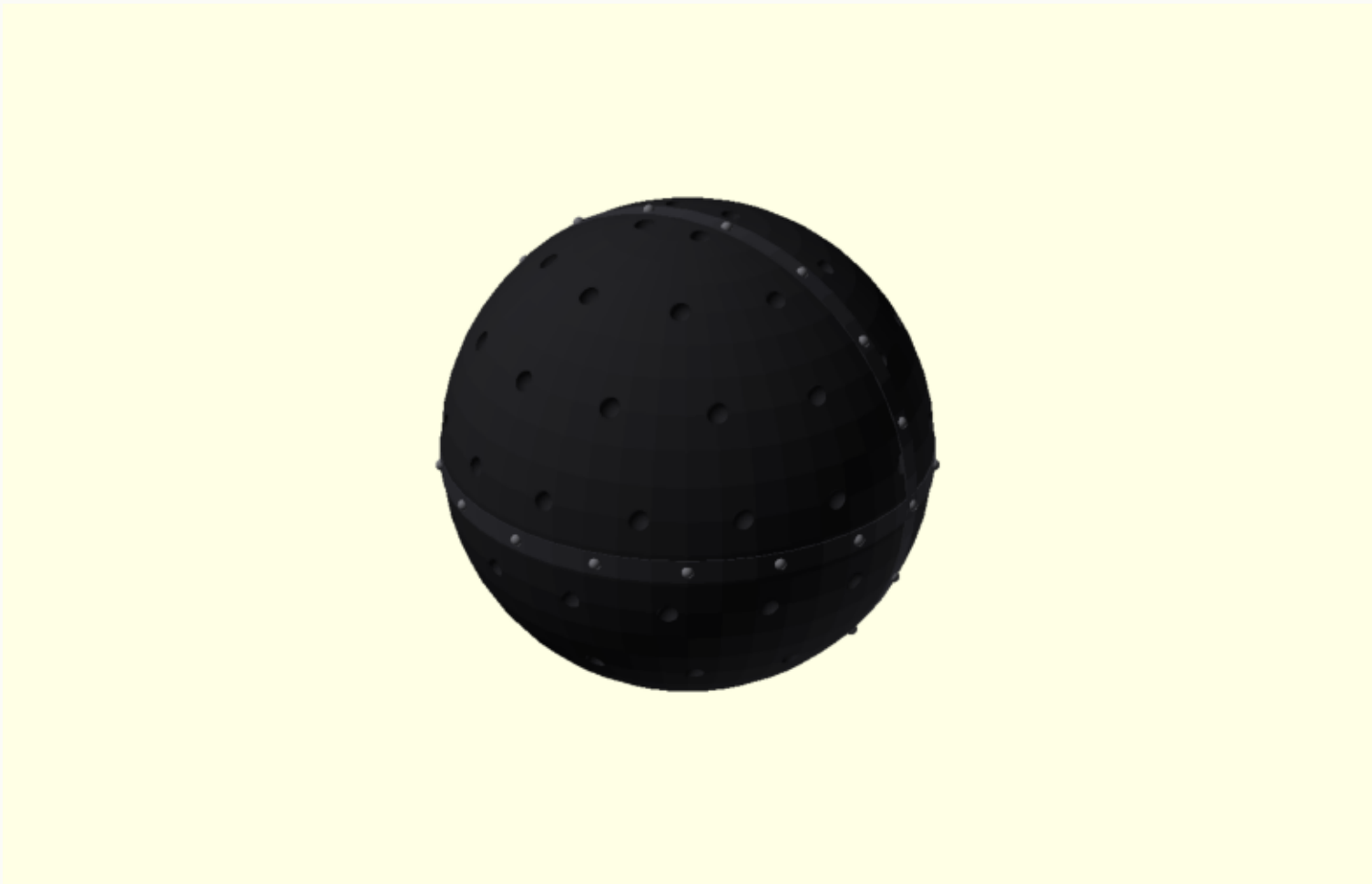
OFFICIAL STORY: it flies by flapping its wings. The wings are thin and genuinely flutter (SNT-02/03/04). The lift is actually the internal micro-EDF swarm (SNT-05).  
Do not explain this to Potterheads.

BILL OF MATERIALS — SNT (14 line items, prototype qty 1)

| REF    | PART                             | SPEC                                          | QTY | \$ EA | \$ EXT |
|--------|----------------------------------|-----------------------------------------------|-----|-------|--------|
| SNT-01 | Gold-clad vented sphere shell    | Ø80 mm (R=40 mm), walnut-sized, dimple vents  | 1   | 90    | 90     |
| SNT-02 | Animatronic flapping wing ass... | thin filigree vane on hinge, span ~76 mm      | 2   | 28    | 56     |
| SNT-03 | Coreless gearmotor + crank       | 6 mm coreless, flap linkage, 8-14 Hz beat     | 2   | 9     | 18     |
| SNT-04 | Wing membrane                    | 5 micron gold-tint mylar, feather-cut         | 2   | 4     | 8      |
| SNT-05 | Micro ducted-fan lift unit       | Ø16 mm internal, the ACTUAL lift              | 4   | 22    | 88     |
| SNT-06 | Micro brushless motor            | 4 mm, paired to micro-EDF                     | 4   | 11    | 44     |
| SNT-07 | Micro ESC (4-in-1)               | 5 A, integrated                               | 1   | 26    | 26     |
| SNT-08 | Nano flight controller           | STM32 class, IMU on-board, predictive evasion | 1   | 38    | 38     |
| SNT-09 | Optical-flow + ToF sensor        | 360deg-ish reach detection                    | 2   | 16    | 32     |
| SNT-10 | Micro LiDAR / proximity ring     | detects incoming hands/airframes              | 1   | 45    | 45     |
| SNT-11 | Capacitive capture sensor        | registers an enclosing hand                   | 1   | 12    | 12     |
| SNT-12 | Micro Li-Po cell                 | 1S 300 mAh, lightweight                       | 1   | 8     | 8      |
| SNT-13 | Gold micro-LED                   | glow so the crowd can track it                | 1   | 5     | 5      |
| SNT-14 | 2.4 GHz telemetry link           | reports pose to the Dementor referee          | 1   | 14    | 14     |

UNIT TOTAL (prototype): \$484

ISOMETRIC



SIDE



TOP



CUTAWAY — internal drive revealed



KEY DIMENSIONS & SPEC

(every number ties back to the flight software)

Shell radius150 mm (Ø300)

ball/params.py bludger.radius = 0.15 m

Mass1.2 kg (foam)

bludger.mass – looks iron, isn't

Top speed15 m/s

bludger.max\_speed

Max accel22 m/s²

bludger.max\_accel (most aggressive)

Lift4 × EDF (internal)

swarm core

Soft-tag radius1.5 m

tag\_radius (outside no-contact floor)

Bat deflect1.6 m

bat\_radius

Contact0 — software hit

ball/avoidance.py, worst clr +0.40 m

NOTE

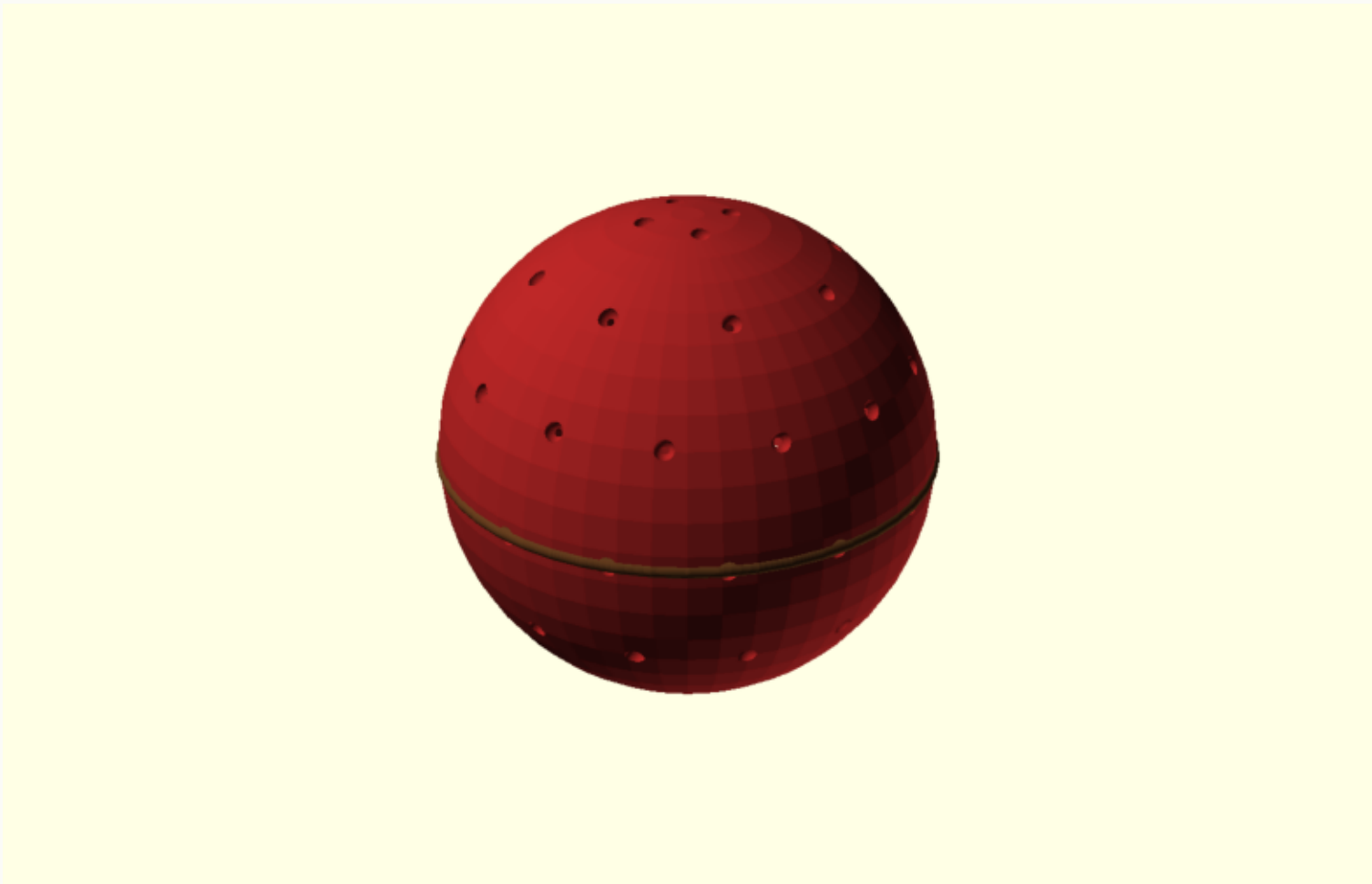
All threat, no contact. The 'hit' is a proximity tag registered in software ~1.5 m out; the foam shell never has to reach a person.

BILL OF MATERIALS — BLG (12 line items, prototype qty 1)

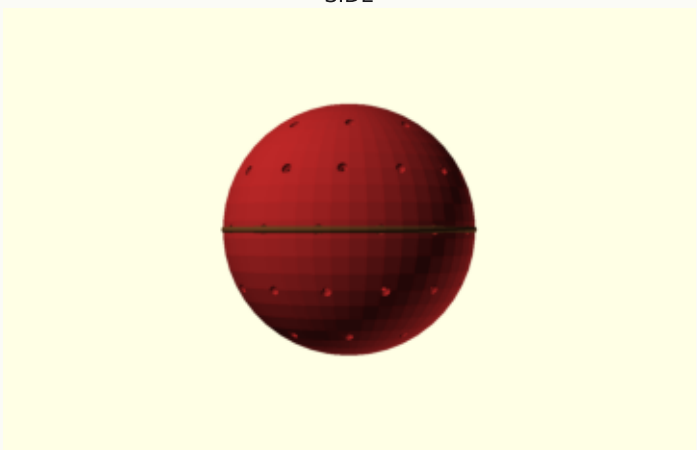
| REF    | PART                           | SPEC                                   | QTY | \$ EA | \$ EXT |
|--------|--------------------------------|----------------------------------------|-----|-------|--------|
| BLG-01 | Black impact-foam sphere shell | Ø300 mm (R=150 mm), riveted-iron look  | 1   | 70    | 70     |
| BLG-02 | Equator rivet bands (cosmetic) | two crossed bands + dummy rivets       | 1   | 18    | 18     |
| BLG-03 | Ducted-fan lift unit           | Ø40 mm internal, internal swarm        | 4   | 40    | 160    |
| BLG-04 | Brushless motor                | per fan, high power-to-weight          | 4   | 22    | 88     |
| BLG-05 | 4-in-1 ESC                     | 20 A, telemetry                        | 1   | 38    | 38     |
| BLG-06 | Flight controller              | IMU on-board, lead-pursuit guidance    | 1   | 42    | 42     |
| BLG-07 | LiDAR / proximity sensor       | target ranging for the soft tag        | 2   | 55    | 110    |
| BLG-08 | Bat-swing detector             | sees the beater's marked bat           | 1   | 16    | 16     |
| BLG-09 | No-contact avoidance compute   | clamps velocity to a stop at the shell | 1   | 0     | 0      |
| BLG-10 | Li-Po pack                     | 3S, sized for sprint pursuits          | 1   | 24    | 24     |
| BLG-11 | Status LED                     | angry red glow                         | 1   | 5     | 5      |
| BLG-12 | Telemetry link                 | reports pose to the Dementor           | 1   | 14    | 14     |

UNIT TOTAL (prototype):\$585

ISOMETRIC



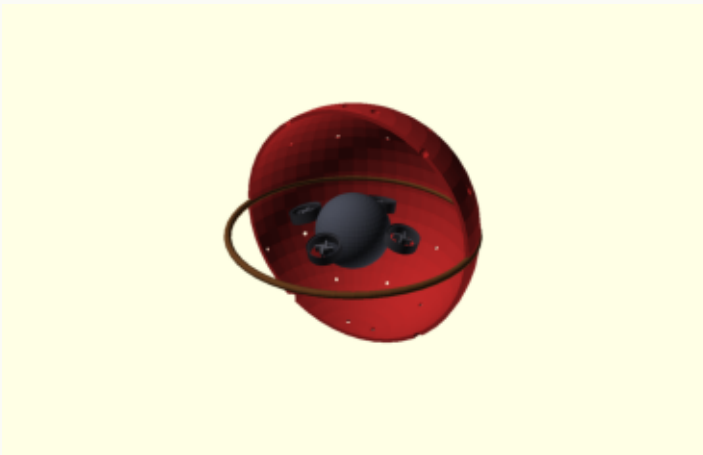
SIDE



TOP



CUTAWAY — internal drive revealed



KEY DIMENSIONS & SPEC

(every number ties back to the flight software)

Shell radius180 mm (Ø360)

ball/params.py quaffle.radius = 0.18 m

Mass0.5 kg

quaffle.mass – light despite its size

Top speed7 m/s

quaffle.max\_speed (gentle)

Catch radius0.45 m

quaffle.catch\_radius (forgiving)

Lift4 × EDF (internal)

swarm core, soft hover

On catchthrust → 0 in hand

ball/quaffle.py capacitive sense

Scoringhoop optical gate

auto pass-through detection

Safetyno-contact floor

even the nice ball won't bonk you

NOTE

Soft hover, generous catch, rides with its holder, then throws to the nearest hoop with tunable aim-assist (on for youth, off for pros).

BILL OF MATERIALS — QAF (12 line items, prototype qty 1)

| REF    | PART                          | SPEC                                          | QTY | \$ EA | \$ EXT |
|--------|-------------------------------|-----------------------------------------------|-----|-------|--------|
| QAF-01 | Red grippy foam sphere shell  | Ø360 mm (R=180 mm), seamed + finger grooves   | 1   | 75    | 75     |
| QAF-02 | Grab-handle ring              | recessed leather ring, holds the catch sensor | 1   | 14    | 14     |
| QAF-03 | Ducted-fan lift unit          | Ø48 mm internal, internal swarm               | 4   | 42    | 168    |
| QAF-04 | Brushless motor               | low-speed, smooth                             | 4   | 20    | 80     |
| QAF-05 | 4-in-1 ESC                    | 15 A, telemetry                               | 1   | 34    | 34     |
| QAF-06 | Flight controller             | IMU on-board, throw + aim-assist              | 1   | 40    | 40     |
| QAF-07 | Capacitive catch sensor array | detects a holding hand -> thrust to 0         | 1   | 18    | 18     |
| QAF-08 | Hoop pass-through beacon      | IR tag read by the hoop optical gate          | 1   | 10    | 10     |
| QAF-09 | No-contact avoidance compute  | even the nice ball won't bonk you             | 1   | 0     | 0      |
| QAF-10 | Li-Po pack                    | 3S, long hover endurance                      | 1   | 22    | 22     |
| QAF-11 | Status LED                    | warm white possession glow                    | 1   | 5     | 5      |
| QAF-12 | Telemetry link                | reports pose to the Dementor                  | 1   | 14    | 14     |

UNIT TOTAL (prototype):\$480

| ITEM | CODE | MATCH QTY | UNIT \$ | EXT \$ |
|------|------|-----------|---------|--------|
|------|------|-----------|---------|--------|

|                 |     |    |          |                  |
|-----------------|-----|----|----------|------------------|
| NIMBUS-9¾ BROOM | BRM | 14 | \$34,303 | <b>\$480,242</b> |
|-----------------|-----|----|----------|------------------|

|         |     |   |       |              |
|---------|-----|---|-------|--------------|
| QUAFFLE | QAF | 1 | \$480 | <b>\$480</b> |
|---------|-----|---|-------|--------------|

|         |     |   |       |                |
|---------|-----|---|-------|----------------|
| BLUDGER | BLG | 2 | \$585 | <b>\$1,170</b> |
|---------|-----|---|-------|----------------|

|               |     |   |       |              |
|---------------|-----|---|-------|--------------|
| GOLDEN SNITCH | SNT | 1 | \$484 | <b>\$484</b> |
|---------------|-----|---|-------|--------------|

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|             |                  |
|-------------|------------------|
| MATCH TOTAL | <b>\$482,376</b> |
|-------------|------------------|